Then ;

$$① \Rightarrow \boxed{u(x) = \left(\frac{b-a}{l}\right)x + a}$$ where ;

PROBLEMS :

Q1 : $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ ——— ①

On comparing with standard eqn : $\alpha^2 = 1$.

i) $u(0,t) = 0$

ii) $u(1,t) = 0$ $\therefore l = 1$.

iii) $u(x,0) = 3\sin n\pi x$, $0 < x < 1$.

Then solution of ① :

$$u(x,t) = (C_1 \cos\lambda x + C_2 \sin\lambda x)\, e^{-\lambda^2 t} \quad ——— ②$$

Put $x = 0$ in ② :

$0 = u(0,t) = C_1 e^{-\lambda^2 t} \Rightarrow C_1 = 0$.

Put $C_1 = 0$ in ② :

$$u(x,t) = C_2 \sin\lambda x \cdot e^{-\lambda^2 t} \quad ——— ③$$

Put $x = 1$ in ③ :

$0 = u(1,t) = C_2 \sin\lambda\, e^{-\lambda^2 t} \Rightarrow \sin\lambda = 0$

$\therefore \lambda = n\pi$; $n = 0, 1, 2, \ldots$

Then ③ $\Rightarrow u(x,t) = C_2 \sin n\pi x\, e^{-\lambda^2 t}$. , $n = 1, 2, 3, \ldots$

The most general solution is :

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin(n\pi x)\, e^{-n^2\pi^2 t} \quad ——— ④$$

Put $t=0$ in (4):

$$u(x,0) = \sum_{n=1}^{\infty} b_n \sin(n\pi x) \quad (1)$$

As given in sin no need to ~~expand~~ (b_n)..

$$3\sin\pi m x = b_1 \sin\pi x + b_2 \sin 2\pi x + \ldots\ldots$$

$\therefore b_m = 3$

$b_n = 0$, if $n \neq m$

(4) $\Rightarrow$ $\boxed{u(x,t) = 3.\sin(m\pi x).e^{-m^2\pi^2 t}}$

Q6: $\dfrac{\partial u}{\partial t} = \alpha^2 . \dfrac{\partial^2 u}{\partial x^2}$ —— (1) , $l = 10$

i) $u(0,t) = 0$ ii) $u(10,t) = 0$

iii) $u(x,0) = f(x) = x(10-x)$; $0 < x < 10$..

i) $\Rightarrow C_1 = 0$ ii) $\Rightarrow \lambda = \dfrac{n\pi}{10}$

Then; $f(x) = u(x,0) = \sum_{n=1}^{\infty} b_n \sin\left(\dfrac{n\pi x}{10}\right)$

As $f(x)$ is not in sin find b_n and use = Sol^n..

$$b_n = \frac{2}{10}\int_0^{10} f(x).\sin\left(\frac{n\pi x}{10}\right)dx = \ldots\ldots$$

Q2. i) $\Rightarrow$ Use $u \to 0$ as $t \to \infty$

Here: i) $\left(\dfrac{\partial u}{\partial x}\right)_{x=0} = 0$ ii) $\left(\dfrac{\partial u}{\partial x}\right)_{x=l} = 0$ (iii) $u(x,0) = f(x) = lx - x^2$.

$0 < x < l$.

i) & ii) implies no heat transfer $\to$ ends insulated

Take;

$$u(x,t) = (C_1\cos\lambda x + C_2 \sin\lambda x).e^{-\lambda^2\alpha^2 t} \quad \text{—— (2)}$$

Diff (2) w.r.t x:

$$\frac{\partial u}{\partial x} = (-C_1\sin\lambda x.\lambda + C_2\lambda\cos\lambda x).e^{-\lambda^2\alpha^2 t} \quad \text{—— (3)}$$

July

Weeks	26	27	28	29	30
Monday	30	2	9	16	23
Tuesday	31	3	10	17	24
Wednesday		4	11	18	25
Thursday		5	12	19	26
Friday		6	13	20	27
Saturday		7	14	21	28
Sunday	1	8	15	22	29

August

Weeks	31	32	33	34	35
Monday		6	13	20	27
Tuesday		7	14	21	28
Wednesday	1	8	15	22	29
Thursday	2	9	16	23	30
Friday	3	10	17	24	31
Saturday	4	11	18	25	
Sunday	5	12	19	26	

Put $x=0$ in (3): $0=\left(\frac{\partial u}{\partial x}\right)_{x=0} = C_2\cdot\lambda\cdot e^{-\alpha^2\lambda^2 t}$. $\Rightarrow C_2=0$

Put $x=l$ in (3): $0=\left(\frac{\partial u}{\partial x}\right)_{x=l} = -C_1\lambda\sin\lambda l\, e^{-\alpha^2\lambda^2 t}$ $\Rightarrow \sin\lambda l=0$

$\Rightarrow \lambda l = n\pi$

$\Rightarrow \lambda=\frac{n\pi}{l}$, $n=0,1,2,\ldots$

(2) $\Rightarrow u(x,t)=C_1\cdot\cos\left(\frac{n\pi x}{l}\right)\cdot e^{-(\alpha^2\pi^2 n^2/l^2)t}$, $n=0,1,\ldots$

The most general solution is:

$$u(x,t)=\sum_{n=0}^{\infty}a_n\cos\left(\frac{n\pi x}{l}\right)e^{-(\alpha^2 n^2\pi^2/l)\cdot t}$$

$$u(x,t)=a_0+\sum_{n=1}^{\infty}a_n\cos\left(\frac{n\pi x}{l}\right)\cdot e^{\left(\frac{-\alpha^2 n^2\pi^2}{l}\right)\cdot t} \quad \text{--- (4)}$$

Put $t=0$:

$$u(x,0)=a_0+\sum_{n=1}^{\infty}a_n\cos\left(\frac{n\pi x}{l}\right)\cdot(1).$$

$$f(x)=\frac{A_0}{2}+\sum_{n=1}^{\infty}a_n\cdot\cos\left(\frac{n\pi x}{l}\right)$$

Then; By Fourier:

$$A_0=\frac{2}{l}\int_0^l f(x)\cdot dx \Rightarrow a_0=\frac{1}{l}\int_0^l f(x)\cdot dx$$

And $a_n=\frac{2}{l}\int_0^l (lx-x^2)\cdot\cos\left(\frac{n\pi x}{l}\right)\cdot dx$

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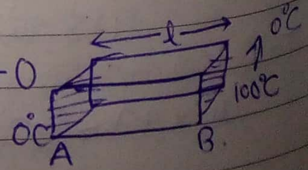
Q3/: (a) Steady State Solⁿ.

$$u(x)=\left(\frac{b-a}{l}\right)x+a=\left(\frac{100-0}{l}\right)x+0$$



i) $u(0,t)=0$ ii) $u(l,t)=0$

iii) $u(x,0)=\frac{100x}{l}$; $0<x<l$.

let (same) $u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{n^2\alpha^2\pi^2}{l^2}t} \rightarrow b_n = \ldots\ldots$

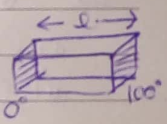
$u(x,t) \to f(x)$

Q5: ii) $u(x,0) = \frac{100x}{l}$, $0 < x, l$ (i) $\left.\frac{\partial u}{\partial x}\right)_{x=0} = 0$ ii) $\left.\frac{\partial u}{\partial x}\right)_{x=l} = 0$

(steady state solⁿ)...

Q3(b): Before Change: $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$ —— (1)

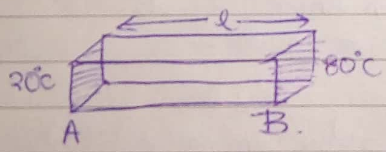
Steady State Solⁿ: $U(x) = \left(\frac{100-0}{l}\right)(x) + 0$



$\therefore U(x) = \frac{100x}{l}$; $0 < x < l$

i) $u(0,t) = 20$ (ii) $u(l,t) = 80$ (iii) $u(x,0) = f(x) = \frac{100x}{l}$

After Change:



Steady State Solⁿ: $u_1(x) = \left(\frac{80-20}{l}\right)x + 20 \Rightarrow u_1(x) = \left(\frac{60}{l}x + 20\right)$

Since, Boundary Cond's are non-zero, we split the solution

$u(x,t) = u_1(x) + v(x,t)$ —— (2)

$u_1(x)$ = steady state soln which again prevails.

$v(x,t)$ = transient solⁿ.

To find: $v(x,t)$.

$v(x,t)$ also satisfies: $\frac{\partial v}{\partial t} = \alpha^2 \frac{\partial^2 v}{\partial x^2}$ —— (3)

(2) $\Rightarrow v(x,t) = u(x,t) - u_1(x)$

Then,

(i) $v(0,t) = u(0,t) - u_1(0) = 20 - 20 = 0$

(ii) $v(l,t) = u(l,t) - u_1(l) = 80 - 80 = 0$

(iii) $v(x,0) = u(x,0) - u_1(x) = \frac{100x}{l} - \frac{60x}{l} - 20 = \frac{40x}{l} - 20$

Now as B Cond's are 0 then;

sol^n of (3)

$\Rightarrow v(x,t) = (C_1 \cos \lambda x + C_2 \sin \lambda x) . e^{-\alpha^2 . \lambda^2 t}$ —— (4)

i) $C_1 = 0$

ii) ~~C_2~~ $\lambda = \frac{n\pi}{l}$; $n = 0, 1, 2 \ldots$

$\Rightarrow v(x,t) = C_2 . \sin\left(\frac{n\pi x}{l}\right) e^{-\frac{\alpha^2 n^2 \pi^2 t}{l^2}}$, $n = 1, 2, 3 \ldots$

$$v(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) . e^{-\frac{\alpha^2 n^2 \pi^2 t}{l^2}} \quad \text{—— (4)}$$

Put $t = 0$

$$v(x,0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \quad (1)$$

$\therefore f(x) = v(x,t)$; $b_n = \frac{2}{l}\int_0^l \left(\frac{40x}{l} - 20\right) \sin\left(\frac{n\pi x}{l}\right) dx$

Then $\Rightarrow v(x,t)$

Replace in (2) :

Q4 {same as 3(b)}

Q7: i) $u(0,t) = 0$

ii) $\left(\frac{\partial u}{\partial x}\right)_{x=l} = A$, A is const. iii) $u(x,0) = 0$ (as throughout)

Since the temp. at end $x = l$ is non-Zero, we split the solution as

We split as: $u(x,t) = u_1(x) + v(x,t)$ ——— (1)

Steady State Sol'n:

$$u_1(x) = C_1 x + C_2 \quad \text{——— (2)}$$

Subject to:

$$u_1(0) = 0 \quad \text{and} \quad \left(\frac{du_1}{dx}\right)_{x=l} = A.$$

Put $x = 0$ in (2):

$$0 = u_1(x) = 0 + C_2 \quad \therefore C_2 = 0.$$

$$\Rightarrow \left(\frac{du_1}{dx}\right)_{x=l} = C_1 \quad \Rightarrow C_1 = A$$

(2) $\Rightarrow$ $u_1(x) = Ax$

(1) $\Rightarrow$ $v(x,t) = u(x,t) - u_1(x)$

Then:

(i) $v(0,t) = u(0,t) - u_1(0) = 0 - 0 = 0$

(ii) $\left(\frac{\partial v}{\partial x}\right)_{x=l} = \left(\frac{\partial u}{\partial x}\right)_{x=l} - \left(\frac{du_1}{dx}\right)_{x=l} = A - A = 0$

(iii) $v(x,0) = u(x,0) - u_1(x) = 0 - Ax = -Ax.$

$$\therefore v(x,t) = (C_1 \cos\lambda x + C_2 \sin\lambda x) \cdot e^{-\alpha^2\lambda^2 t} \quad \text{——— (3)}$$

i) $\Rightarrow C_1 = 0$

Then;

$$v(x,t) = C_2 \sin\lambda x \cdot e^{-\alpha^2\lambda^2 t} \quad \text{——— (4)}$$

Diff w.r.t x:

$$\left(\frac{\partial v(x,t)}{\partial x}\right) = C_2 \cdot \lambda \cdot \cos\lambda x \cdot e^{-\alpha^2\lambda^2 t}$$

At $x = l \Rightarrow C_2 \lambda \cos\lambda l \; e^{-\alpha^2\lambda^2 t} = 0$

if: $\cos\lambda l = 0 \Rightarrow \lambda l = \frac{(2n-1)\pi}{2}$

$$\therefore \lambda = \frac{(2n-1)\pi}{2l}$$

$$④ \Rightarrow v(x,t) = C_2 \cdot \sin\left(\frac{(2n-1)\pi}{2l}\right)x \cdot e^{-\frac{\alpha^2(2n-1)^2\pi^2}{4l^2}\cdot t}, \quad n = 1,2,3\ldots$$

The most general solution is:

$$v(x,t) = \sum_{n=1}^{\infty} b_n \sin\left[\frac{(2n-1)\pi}{2l}\right]x \cdot e^{-\frac{\alpha^2(2n-1)^2\pi^2}{4l^2}\cdot t} \quad \text{——} ⑤$$

$\therefore$ Put $t = 0$:

$$f(x) = v(x,0) = \sum_{n=1}^{\infty} b_n \sin\left[\frac{(2n-1)\pi}{2l}\right]x$$

$$\therefore b_n = -\frac{2A}{l}\int_0^l x \cdot \sin\left[\frac{(2n-1)\pi}{2l}\right]\cdot x$$

$$= -\frac{2A}{l}\left[x \cdot \frac{-\cos\left(\frac{(2n-1)\pi x}{2l}\right)}{\frac{(2n-1)\pi}{2l}} + (1)\cdot\frac{\sin\left(\frac{(2n-1)\pi x}{2l}\right)}{\left(\frac{(2n-1)^2\pi^2}{(2l)^2}\right)}\right]_0^l$$

$$= -\frac{2A}{l}\left[\frac{-2l^2}{(2n-1)\pi}\cdot\cos\left(\frac{(2n-1)\pi}{2}\right) + \frac{4l^2}{(2n-1)^2\pi^2}\cdot\sin\left(\frac{(2n-1)\pi}{2}\right)\right]$$

$$\cos\left(\frac{(2n-1)\pi}{2}\right) = \cos\left(n\pi - \frac{\pi}{2}\right) = \cos n\pi \cdot \cos\frac{\pi}{2} + \sin n\pi \cdot \sin\frac{\pi}{2} = 0.$$

$$\sin\left(\frac{(2n-1)\pi}{2}\right) = \sin\left(n\pi - \frac{\pi}{2}\right) = \sin n\pi \cdot \cos\frac{\pi}{2} + \cos n\pi \cdot \sin\frac{\pi}{2} = -(-1)^n$$

$$\therefore b_n = +\frac{2A}{l}\left[\frac{4l^2}{(2n-1)^2\pi^2}(-1)^n\right] = \frac{8Al(-1)^n}{(2n-1)^2\cdot\pi^2}$$

$$⑤ \Rightarrow v(x,t) = \frac{8Al}{\pi^2}\sum_{n=1}^{\infty}\frac{(-1)^n}{(2n-1)^2}\cdot\sin\left[\frac{(2n-1)\pi x}{2l}\right]\cdot e^{\left[-\frac{(2n-1)^2\pi^2\alpha^2 t}{4l^2}\right]}$$

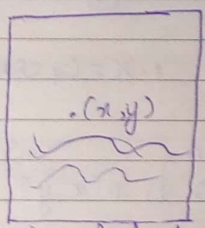
September					
Weeks	35	36	37	38	39
Monday		3	10	17	24
Tuesday		4	11	18	25
Wednesday		5	12	19	26
Thursday		6	13	20	27
Friday		7	14	21	28
Saturday	1	8	15	22	29
Sunday	2	9	16	23	30

October					
Weeks	40	41	42	43	44
Monday	1	8	15	22	29
Tuesday	2	9	16	23	30
Wednesday	3	10	17	24	31
Thursday	4	11	18	25	
Friday	5	12	19	26	
Saturday	6	13	20	27	
Sunday	7	14	21	28	

TWO DIMENSIONAL HEAT EQUATION:

(Take sheet):

$$\frac{\partial u}{\partial t} = \alpha^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$



Here ; $u(x,y,t)$

Back & Front side of metal sheet are thermally insulated.

Not t, no initial cond's.

TWO DIMENSION HEAT EQ^N IN STEADY STATE

$$\frac{\partial u}{\partial t} = 0 \quad ; \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{—(1)}$$

(Same as Laplace Eqn in 2D)

$$u(x,y) = X(x).Y(y) = X.Y \quad \text{—(2)}$$

$$\frac{\partial^2 u}{\partial x^2} = X''.Y \quad ; \quad \frac{\partial^2 u}{\partial y^2} = XY''$$

(1) $\Rightarrow$ $X''Y + XY'' = 0$

$X''Y = -XY''$

$\Rightarrow \dfrac{X''}{X} = \dfrac{-Y''}{Y} = k$ (constant). —(3) $\Rightarrow X'' - kX = 0$ —(3)

$Y'' + kY = 0$ —(4)

Case–I: If $k=0$:

(3) $\Rightarrow$ $X''=0$

$X' = c_1$

$X = c_1 x + c_2$

(4) $\Rightarrow$ $Y''=0$

$Y' = c_3$

$Y = c_3 y + c_4$.

Case–II: If $k = \lambda^2 > 0$

(3) $\Rightarrow$ $X'' - \lambda^2 X = 0$

A.E : $m = \pm\lambda$

$X = c_5 e^{\lambda x} + c_6 e^{-\lambda x}$

(4) $\Rightarrow$ $Y'' + \lambda^2 Y = 0$

A.E. $\Rightarrow m = 0 \pm i\lambda$

$Y = c_7 \cos\lambda y + c_8 \sin\lambda y$.

July

Weeks	26	27	28	29	30
Monday	30	2	9	16	23
Tuesday	31	3	10	17	24
Wednesday		4	11	18	25
Thursday		5	12	19	26
Friday		6	13	20	27
Saturday		7	14	21	28
Sunday	1	8	15	22	29

Case – III: If $k = -\lambda^2 < 0$

(3) ⇒ $x'' + \cancel{k}\lambda^2 = 0$ (4) ⇒ $Y'' - \lambda^2 Y = 0$

∴ $X = C_9 \cos\lambda x + C_{10} \sin\lambda x$ $Y = C_{11} e^{\lambda y} + C_{12} e^{-\lambda y}$

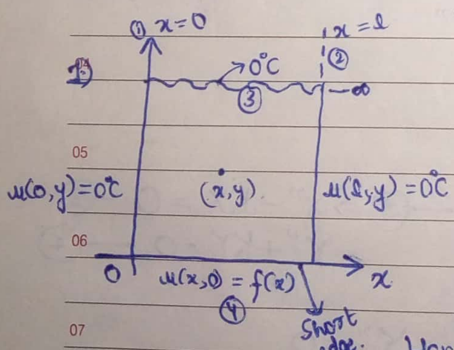
Then

(2) ⇒ i) $u(x,y) = (C_1 x + C_2)(C_3 y + C_4)$

ii) $u(x,y) = (C_5 e^{\lambda x} + C_6 e^{-\lambda x})(C_7 \cos\lambda y + C_8 \sin\lambda y)$

iii) $u(x,y) = (C_9 \cos\lambda x + C_{10} \sin\lambda x)(C_{11} e^{\lambda y} + C_{12} e^{-\lambda y})$

TEMP. DISTRIBUTION IN LONG PLATE (Infinite Plate)



i) $u(0,y) = 0$
ii) $u(l,y) = 0$
iii) $u(x,\infty) = 0$
(iv) $u(x,0) = f(x)$

Here 1) tends to zero 2) also

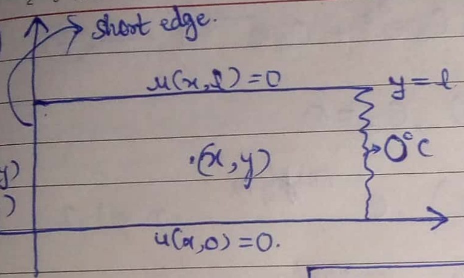
12 Sun

Then;

$$\boxed{u(x,y) = (C_1 \cos\lambda x + C_2 \sin\lambda x)(C_3 e^{\lambda y} + C_4 e^{-\lambda y})}$$

Note: If extended indefinitely in y then y has e^x term and x trigo term

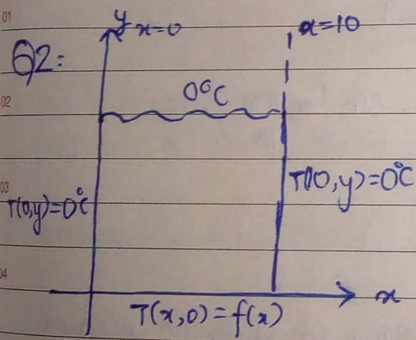
2.)



The suitable solution : $\boxed{u(x,y) = (C_1 e^{\lambda x} + C_2 e^{-\lambda x})(C_3 \cos\lambda y + C_4 \sin\lambda y)}$

PROBLEMS:

Q2:



i) $T(0,y)=0$

ii) $T(10,y)=0$

iii) $T(x,\infty)=0$

iv) $T(x,0)=f(x) = 4(10x-x^2)$; $0<x<10.$

The $T(x,y)$ satisfies $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$ —— (1)

The most suitable solution of (1) is :

$$T(x,y) = (C_1\cos\lambda x + C_2 \sin\lambda x)(C_3 e^{\lambda y} + C_4 e^{-\lambda y}) \text{ —— (2)}$$

Put $x=0$ in (2):

$0 = T(0,y) = C_1(C_3 e^{\lambda y} + C_4 e^{-\lambda y})$ $\Rightarrow \boxed{C_1 = 0}$

(1) $\Rightarrow T(x,y) = C_2 \sin\lambda x\,(C_3 e^{\lambda y} + C_4 e^{-\lambda y})$ —— (3)

Put $x=0$ in (3)

$0 = T(10,y) = C_2 \sin 10\lambda\,(C_3 e^{\lambda y} + C_4 e^{-\lambda y})$ $\Rightarrow \sin 10\lambda = 0$

$\Rightarrow 10\lambda = n\pi$, $\lambda = \frac{n\pi}{10}$

$$③ \Rightarrow T(x,y) = C_2 \sin\left(\frac{n\pi x}{10}\right)\left[C_3 e^{\lambda y} + C_4 e^{-\lambda y}\right] \quad \text{——} ④$$

In order to satisfy condition (iii), $C_3 = 0$

$$④ \Rightarrow T(x,y) = C_2 \cdot C_4 \sin\left(\frac{n\pi x}{10}\right) \cdot e^{-n\pi y/10}, \quad n = 1, 2 \ldots$$

$$T(x,y) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{10}\right) \cdot e^{-n\pi y/10} \quad \text{——} ⑤$$

Put $y = 0$:

$$f(x) = T(x,0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{10}\right) \cdot 1$$

By half range: $b_n = \frac{2}{10}\int_0^{10} (10x - x^2) \cdot \sin\left(\frac{n\pi x}{10}\right) dx$

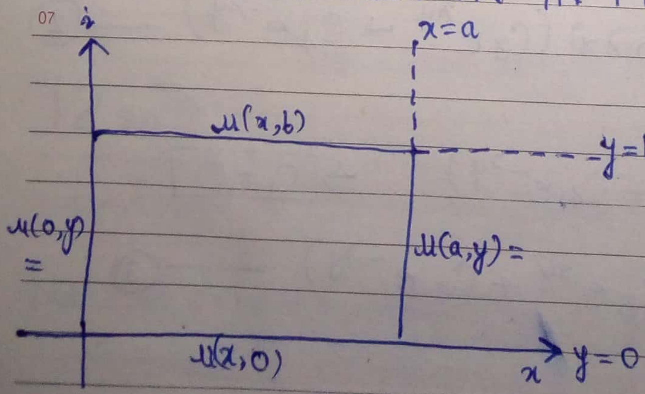
↳ Use in ⑤:

Q3 Same type (but y): 4, 5, 6, 7, 8, 9, 10, 11, 12, 1.
x present.

Note: If f(x) ie condition of Q in terms of sine no need of b_n if in x then use..

~~Q1~~

TEMP. DISTRIBUTION IN FINITE PLATES:



So The suitable solution is:

$$u(x,y) = (C_1 \cos \lambda x + C_2 \sin \lambda x)(C_3 e^{\lambda y} + C_4 e^{-\lambda y})$$

(Or)

$u(x,y) =$

(Or):

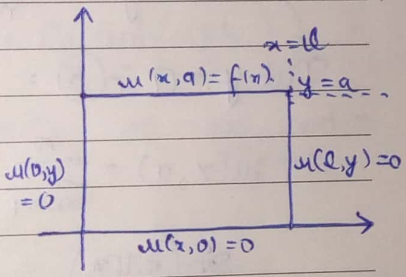
$$\boxed{u(x,y) = (C_1 \cos\lambda x + C_2 \sin\lambda x)(C_3 \cosh\lambda y + C_4 \sinh\lambda y)}$$

---- For temp. in y (above and top)
(Rev. for x).

PROBLEM:

Q1: Given; $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ —— (1)

i) $u(0,y) = 0$
ii) $u(l,y) = 0$ } Take first lly zero ones.
iii) $u(x,0) = 0$
iv) $u(x,a) = f(x) = \sin\left(\dfrac{n\pi x}{l}\right)$; $0 < x < l$.



Take: $u(x,y) = (c_1 \cos\lambda x + c_2 \sin\lambda x)(C_3 \cosh\lambda y + C_4 \sinh\lambda y)$ —— (2)

Put $x=0$ in (2):

$0 = u(0,y) = C_1(C_3 \cosh\lambda y + C_4 \sinh\lambda y)$ (by (i)).

$\therefore C_1 = 0$

(2) $\Rightarrow u(x,y) = C_2 \sin\lambda x\,(C_3 \cosh\lambda y + C_4 \sinh\lambda y)$ —— (3)

Put $x = l$ in (3):

$0 = u(l,y) = C_2 \sin\lambda l\,(C_3 \cosh\lambda y + C_4 \sinh\lambda y)$

$\sin\lambda l = 0$

$\Rightarrow \lambda = \dfrac{m\pi}{l}$, $m = 0, 1, 2, \ldots$

(3) $\Rightarrow u(x,y) = c_2 \sin\left(\dfrac{m\pi x}{l}\right)\left[C_3 \cosh\left(\dfrac{m\pi y}{l}\right) + C_4 \sinh\left(\dfrac{m\pi y}{l}\right)\right]$ —— (4)

Put $y = 0$ in (4):

$$u(x,0) = C_2 \sin\left(\frac{m\pi x}{l}\right) \cdot C_3 \Rightarrow \boxed{C_3 = 0}$$

$$④ \Rightarrow u(x,y) = C_2 \cdot C_4 \cdot \sin\left(\frac{m\pi x}{l}\right) \sinh\left(\frac{m\pi y}{l}\right) \quad ; m = 1,2,3\ldots$$

The most general sol^n is
↓ Changes

$$u(x,y) = \sum_{m=1}^{\infty} b_m \sin\left(\frac{m\pi x}{l}\right) \cdot \sinh\left(\frac{m\pi y}{l}\right) \quad — ⑤$$

Put $y = a$ in ⑤ :

$$u(x,a) = \sum_{m=1}^{\infty} b_m \cdot \sin\left(\frac{m\pi x}{l}\right) \cdot \sinh\left(\frac{m\pi a}{l}\right)$$

$$\sin\left(\frac{n\pi x}{l}\right) = \sum_{m=1}^{\infty} B_m \cdot \sin\left(\frac{m\pi x}{l}\right)$$

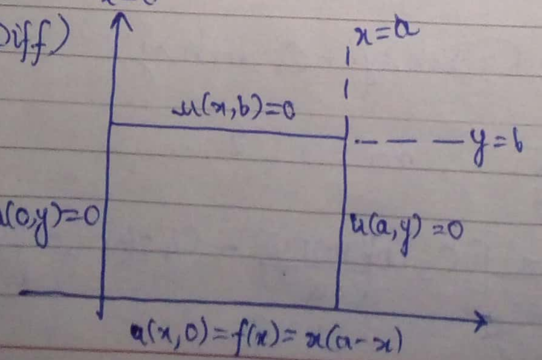
$$\therefore B_n = 1 \text{ and } B_m = 0, \text{ if } m \neq n \quad \left\{ B_m = b_m \cdot \sinh\left(\frac{m\pi a}{l}\right) \right.$$

$$\therefore b_n = \frac{1}{\sinh\left(\frac{n\pi a}{l}\right)} = \operatorname{cosech}\left(\frac{n\pi a}{l}\right)$$

$$b_m = 0, \text{ if } m \neq n.$$

$$⑤ \Rightarrow u(x,y) = \operatorname{cosech}\left(\frac{n\pi a}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot \sinh\left(\frac{n\pi y}{l}\right).$$

Q2 (Diff)



↑ As at axis then eqn in 3rd step.

i) $u(0,y) = 0$
ii) $u(a,y) = 0$
iii) $u(x,b) = 0$
iv) $u(x,0) = f(x) = x(a-x)$; $0 < x < a$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad — ①$$

Here; $u(x,y) = (C_1 \cos\lambda x + C_2 \sin\lambda x)(C_3 \cosh\lambda y + C_4 \sinh\lambda y)$ —②

i) $\Rightarrow C_1 = 0$

ii) $\Rightarrow \lambda = (n\pi/a)$

③ $\Rightarrow u(x,y) = C_2 \sin\left(\frac{n\pi x}{a}\right)\left(C_3 \cosh\left(\frac{n\pi y}{a}\right) + C_4 \sinh\left(\frac{n\pi y}{a}\right)\right)$ —④

Put $y = b$ in ④:

$0 = u(x,b) = C_2 \sin\left(\frac{n\pi x}{a}\right)\left(C_3 \cosh\left(\frac{n\pi b}{a}\right) + C_4 \sinh\left(\frac{n\pi b}{a}\right)\right)$

$\Rightarrow C_3 \cosh\left(\frac{n\pi b}{a}\right) + C_4 \sinh\left(\frac{n\pi b}{a}\right) = 0$ ⟵ Not zero as ④ has y..

$\therefore C_4 = -C_3 \dfrac{\cosh\left(\frac{n\pi b}{a}\right)}{\sinh\left(\frac{n\pi b}{a}\right)}$

Then; ④ $\Rightarrow u(x,y) = C_2 \sin\left(\frac{n\pi x}{a}\right) . C_3 \left\{\cosh\left(\frac{n\pi y}{a}\right) - \dfrac{\cosh\left(\frac{n\pi b}{a}\right)}{\sinh\left(\frac{n\pi b}{a}\right)} . \sinh\left(\frac{n\pi y}{a}\right)\right\}$

$= C_2 . C_3 \sin\left(\frac{n\pi x}{a}\right)\left\{\dfrac{\cosh\left(\frac{n\pi y}{a}\right) . \sinh\left(\frac{n\pi b}{a}\right) - \cosh\left(\frac{n\pi b}{a}\right) . \sinh\left(\frac{n\pi y}{a}\right)}{\sinh\left(\frac{n\pi b}{a}\right)}\right\}$

Note: ~~$= C_2 . C_3$.~~

Relationship b/w trigo & hyperbolic:

$\Rightarrow \sin iA = i \sinh A$

$\Rightarrow \cos iA = \cosh A$

And:

$A \longrightarrow iA$ & $B \longrightarrow iB$.

$\sinh(A \pm B) = (\sinh A . \cosh B \pm \cosh A . \sinh B)$

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July					
Weeks	26	27	28	29	30
Monday	30	2	9	16	23
Tuesday	31	3	10	17	24
Wednesday		4	11	18	25
Thursday		5	12	19	26
Friday		6	13	20	27
Saturday		7	14	21	28
Sunday	1	8	15	22	29

August					
Weeks	31	32	33	34	35
Monday		6	13	20	27
Tuesday		7	14	21	28
Wednesday	1	8	15	22	29
Thursday	2	9	16	23	30
Friday	3	10	17	24	31
Saturday	4	11	18	25	
Sunday	5	12	19	26	

And : $\cos(A \pm B) = \cos A . \cos B \mp \sin A . \sin B$.

$\cos i(A+B) \Rightarrow \cosh(A \pm B) = \cosh A . \cosh B \pm \sinh A . \sinh B$

Then ;

$$u(x,y) = C_2 . C_3 \operatorname{cosech}\left(\frac{n\pi b}{a}\right) . \sin\left(\frac{n\pi x}{a}\right) \left[\sinh\left(\frac{n\pi b}{a} - \frac{n\pi y}{a}\right)\right]$$

$$= C_2 . C_3 \operatorname{cosech}\left(\frac{n\pi b}{a}\right) . \sin\left(\frac{n\pi x}{a}\right) . \sinh\left[\frac{n\pi}{a}(b-y)\right] \quad n = 1, 2, 3 \ldots$$

Now, take general solution:

$$u(x,y) = \sum_{n=1}^{\infty} b_n . \operatorname{cosech}\left(\frac{n\pi b}{a}\right) . \sin\left(\frac{n\pi x}{a}\right) . \sinh\left(\frac{n\pi}{a}(b-y)\right) \quad \text{—(5)}$$

Put $y = 0$:

$$f(x) = \sum_{n=1}^{\infty} b_n \cancel{sr} \frac{1}{\sinh\left(\frac{n\pi b}{a}\right)} . \sin\left(\frac{n\pi x}{a}\right) . \sin\left(\frac{n\pi b}{a}\right)$$

$$= \sum_{n=1}^{\infty} b_n . \sin\left(\frac{n\pi x}{a}\right) \quad , 0 < x < a .$$

$$\therefore b_n = \frac{2}{a}\int_0^a (ax - x^2) . \sin\left(\frac{n\pi x}{a}\right) dx \leftarrow \text{Half range}$$

$$= \frac{2}{a}\left[(ax - x^2)\left\{\frac{-\cos\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)}\right\} - (a - 2x)\left\{\frac{-\sin\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)^2}\right\} + \cancel{a}(-2)\left\{\frac{\cos\left(\frac{n\pi x}{a}\right)}{\left(\frac{n\pi}{a}\right)^3}\right\}\right]_0^a$$

19 Sun

Q8 note : lower edge $u(x, 0) = 100°C$. and find $u\left(\frac{a}{2}, \frac{a}{2}\right)$.